

ESE_ December2020_COMP_Soft Computing_ UCEE702_ Sem VII_ Dec 30,2020_ 10:30_ Objective Questions (20 Marks)

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Questions

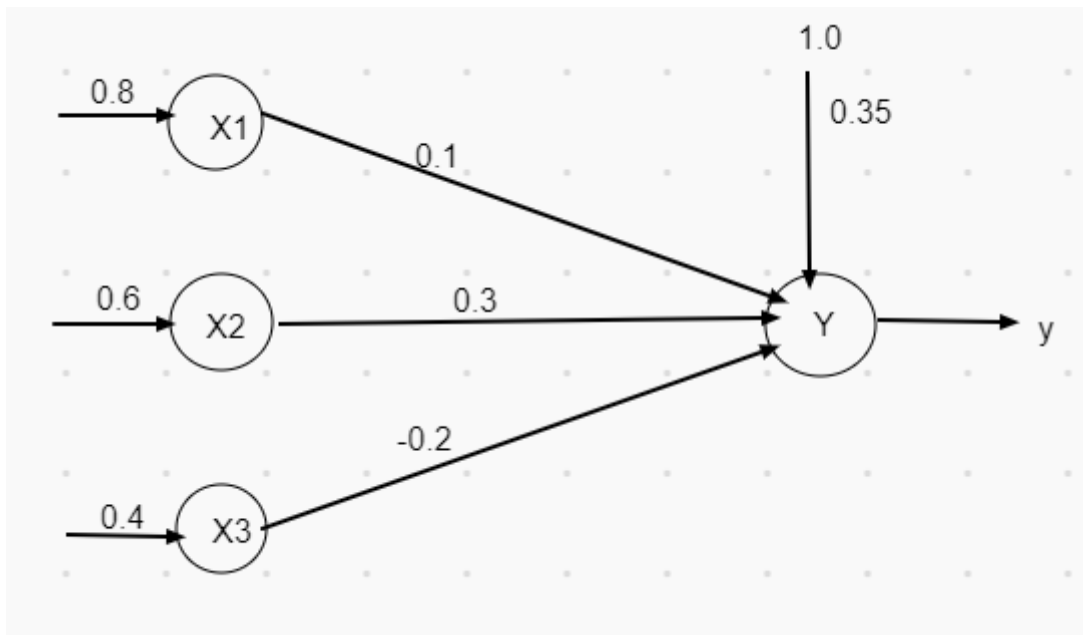
4. Soft computing deals with :

1 point

Mark only one oval.

- ☐ Imprecision , Uncertainty , Partial truth and Approximation
- ☐ Deterministic, certain, complete and precise information
- ☐ Computer generated data, Big data, analytics
- ☐ pure conceptual algorithms, processes, system calls

5. The output of the neuron y for the given network using Binary sigmoidal activation function is: 2 points



Mark only one oval.

- ☐ 0.625
- ☐ 0.375
- ☐ -0.625
- ☐ -0.75

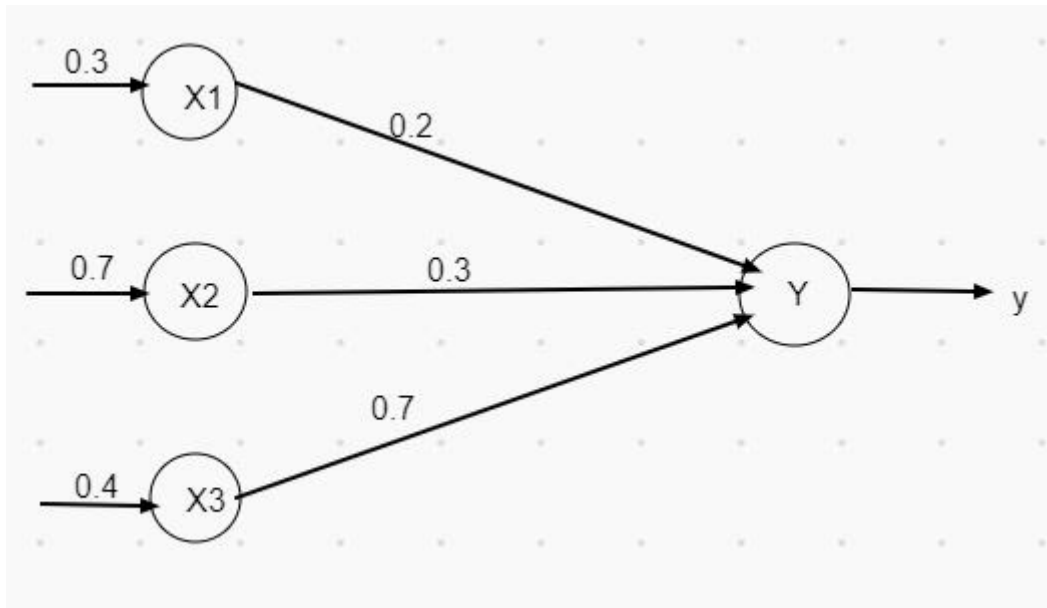
6. An artificial neural network : 1 point

Mark only one oval.

- ☐ updates connecting weights, focus on change in the network
- ☐ updates connecting weights
- ☐ focuses on change in network
- ☐ updates bias value

7. For the given network, the net input at neuron y is:

2 points



Mark only one oval.

☐ 0.65

☐ 0.55

☐ 0.35

☐ 0.45

8. Match the pairs: a. Cell, b. Dendrites, c. Soma, d. Axon 1. Output, 2. Net input, 3. Neuron, 4. Weights or interconnections

1 point

Mark only one oval.

☐ a-4, b-3, c-1, d-2

☐ a-3, b-4, c-2, d-1

☐ a-1, b-2, c-4, d-3

☐ a-2, b-1, c-4, d-3

9. McCulloch-Pit's neuron model can accept input in _____ form(s). 1 point

Mark only one oval.

- ☐ Integer
- ☐ Bipolar
- ☐ Boolean
- ☐ Electrical signals

10. The _____'s law states that: If two neurons on either side of a connection are activated synchronously, then the weight of that connection is increased. 1 point

Mark only one oval.

- ☐ Boltzman
- ☐ McCulloch-Pit
- ☐ Bool
- ☐ Hebb

11. Which of the following can be used for data clustering? 1 point

Mark only one oval.

- ☐ Single layer perception
- ☐ Multilayer perception
- ☐ Self organizing map
- ☐ Radial basis function

12. In a neural network, the number of nodes in the input layer is 10 and the hidden layer is 5. The maximum number of connections from the input layer to the hidden layer would be- 1 point

Mark only one oval.

- ☐ 50
- ☐ less than 50
- ☐ More than 50
- ☐ It is an arbitrary value

13. A neural network without an activation function is essentially just _____ 1 point

Mark only one oval.

- ☐ a linear regression model
- ☐ a Hebb network
- ☐ a Single layer perceptron
- ☐ not a neural network at all

14. In a dishwasher, the fuzzy logic would consider these parameters to set the washing cycle: 1 point

Mark only one oval.

- ☐ number of dishes and greasiness on the dishes
- ☐ amount of leftover food on dishes
- ☐ number of dishes, greasiness and the amount of leftover food served on the dishes.
- ☐ atmospheric temprature, number of dishes and the amount of leftover food served on the dishes.

15. In a neural network that is supposed to identify handwritten numbers(e.g. reading handwritten pincode), the number of neurons in the output layer should be: 1 point

Mark only one oval.

- ☐ 2
- ☐ 1
- ☐ 10
- ☐ 9

16. Fill in the blanks. In the Kohonen's network, a neuron learns by shifting its weights from _____ connections to _____ones 1 point

Mark only one oval.

- ☐ active, inactive
- ☐ inactive, active
- ☐ bias, output error signal
- ☐ output error signal, bias

17. Learning vector quantization is a special case of_____ 1 point

Mark only one oval.

- ☐ Error correction learning
- ☐ Memory based learning
- ☐ Competitive learning
- ☐ Reinforcement learning

18. If A and B are two fuzzy sets given as: $A = \{(x_1, 0.5), (x_2, 0.7), (x_3, 0)\}$ $B = \{(x_1, 0.8), (x_2, 0.2), (x_3, 1)\}$, then $A \cup B$ would be: 1 point

Mark only one oval.

- ☐ $\{(x_1, 0.5), (x_2, 0.2), (x_3, 0)\}$
- ☐ $\{(x_1, 0.8), (x_2, 0.7), (x_3, 1)\}$
- ☐ $\{(x_1, 0.40), (x_2, 0.14), (x_3, 0)\}$
- ☐ $\{(x_1, 0.13), (x_2, 0.9), (x_3, 1)\}$

19. Given fuzzy set $\tilde{A} = \{0.1/1 + 0.3/2 + 0.8/3 + 1/4 + 1/5 + 0.8/6\}$, Core of $\tilde{A}$ would be: 1 point

Mark only one oval.

- ☐ $\{1, 2, 3, 4, 5, 6\}$
- ☐ $\{2, 3, 4\}$
- ☐ $\{5\}$
- ☐ $\{4, 5\}$

20. The minimum number of layers Deep learning networks are: 1 point

Mark only one oval.

- ☐ 2
- ☐ 3
- ☐ 4
- ☐ 1

21. A: It is possible to train a network well by initializing all the weights as 0. B: It is possible to train a network well by initializing biases as 0. Which of these statements is/ are true? 1 point

Mark only one oval.

- ☐ A is true, B is false
- ☐ B is true, A is false
- ☐ A & B are true
- ☐ A & B are false

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